

Critical Speed and Resonance Criteria of Railway Bridge Response to Moving Trains

Lei Mao¹ and Yong Lu, F.ASCE²

Abstract: The dynamic response of railway bridges to moving trains is complicated because of the involvement of the moving loads and the moving masses. Among various response characteristics, bridge resonance is of particular interest in terms of the structural effect and safety of the bridge. As far as the global bridge response is concerned, it is generally understood that when one of the apparent trainload excitation frequencies coincides with the fundamental natural frequency of the bridge, resonance could occur. However, such a general criterion is of little practical use because a typical trainload would involve numerous apparent frequencies (at equal intervals); consequently, for a given bridge (natural frequency), there could be many train speeds that satisfy the preceding resonance condition. Therefore, it is necessary to establish the relative severity of the resonance associated with each resonance scenario. This paper presents the development of a new resonance severity indicator, called Z factor, for the assessment of the resonance effect. It is found that the resonance severity is essentially governed by the ratio between the bridge and carriage lengths. When the carriage mass is significant, the same Z factor will apply; however, the underlying resonance speeds will change because of the altered natural frequency of the bridge-train system. Numerical results demonstrate that the proposed methods are effective for the determination of the resonance effects associated with the potential resonance speeds. DOI: 10.1061/(ASCE)BE.1943-5592.0000336. © 2013 American Society of Civil Engineers.

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Introduction

Railway bridge response to trainload excitation is a major subject in the bridge dynamics and railway engineering fields. Early research can be traced back to the nineteenth century, with the investigation of the collapse of the Chaster Rail Bridge in England in 1847 (Stokes 1849). Generally speaking, as suggested by Yang et al. (2004) in a comprehensive review of the bridge-train/vehicle interaction studies, before the 1940s, the investigations on bridge dynamic responses were performed mainly with analytical or approximate solutions for simple and fundamental problems. After that, analysis with more realistic bridge and vehicle models became possible with the advent of digital computers. Recently, there is a tendency of increasingly resorting to more sophisticated vehicle-bridge coupled models, which can provide insight into more detailed bridge and vehicle responses (Majka and Hartnett 2008; Kwark et al. 2004; Dinh et al. 2009; Liu et al. 2009); nevertheless, relatively simple models that accommodate basic features as moving load and moving mass still prove to be effective for the purpose of identifying key parameters that govern the bridge dynamic response, including the bridge resonance phenomenon (Li and Su 1999; Yau 2001; Garinei and Risitano 2008; Martinez-Rodrigo et al. 2010).

As far as the maximum bridge response under moving vehicles is concerned, a basic scenario that has been studied extensively is the bridge response under a single moving load. The global response of the bridge, which may be represented by the midspan deflection, is known to vary as the speed of the moving load changes, and a critical speed exists under which the bridge experiences the maximum response. The critical speed has been found to closely relate to the fundamental natural frequency of the bridge (Fryba 1999; Yang et al. 2004), and a general relationship is available for cases where the moving mass is negligible. However, such a relationship does not cover the situation where the moving mass itself is large with respect to the bridge mass, as often is the case in relatively small bridges under a heavily laden truck or carriage. Studies (Dehestani et al. 2009) have shown that the involvement of the vehicle mass generally reduces the critical speed, but no specific relationship has been established.

The bridge response under a moving train is more complicated because the frequency contents in the excitation involve not only characteristics from a moving load but also repeated load pulses from consecutive carriages. Existing studies on this general subject may be loosely divided into two groups: one concerns the variation of the natural frequencies of the responding bridge during the passage of a laden train/vehicle (Li and Su 1999; Li et al. 2003; Yang and Chang 2009), and the other deals with the frequency contents in the trainload excitation and their general effect in the bridge response (Paultre et al. 1995; Fryba 2001; Yang and Lin 2005; Ju et al. 2009). In particular, it has been demonstrated that the primary frequencies in the bridge response may be attributable to (1) the so-called driving frequencies that are associated with the duration of a vehicle passing over the bridge (Yang and Lin 2005), and (2) the so-called dominant frequencies arising from the repeated loads and hence being related to the time interval between two consecutive carriages (Paultre et al. 1995; Fryba 2001; Yang and Lin 2005; Ju et al. 2009). Despite

¹Ph.D. Student, Institute for Infrastructure and Environment, School of Engineering, Univ. of Edinburgh, Edinburgh EH9 3JL, U.K.

²Professor, Institute for Infrastructure and Environment, School of Engineering, Univ. of Edinburgh, Edinburgh EH9 3JL, U.K. (corresponding author). E-mail: yong.lu@ed.ac.uk

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the identification of these frequency factors, however, the understanding of their significance in the bridge response remains rather general and qualitative.

This paper presents a study on the resonance phenomenon in the railway bridge response. Emphasis is placed on the evaluation of resonance from the perspective of the key frequencies involved, especially the apparent frequencies, including the aforementioned driving and dominant frequencies in the moving excitation loads. The influence of the moving mass is explicitly considered to incorporate the variation of the dynamic properties of the coupled system. To facilitate the examination of the frequency relationships, it is found that the relative length of the bridge with respect to the length of the carriage is a vital parameter, and this length ratio is adopted in the classification of the frequency response characteristics. To simplify the analysis while maintaining the representativeness of the primary frequency characteristics concerning the bridge response, the vehicle dynamics and track irregularity are omitted in this study.

Overview of the Frequency Characteristics of Moving Load Excitation

The dynamic equation for a bridge coupled with a moving vehicle may be represented by the following classical beam equation:

$$m_b \frac{\partial^2 w_b}{\partial t^2} + EI \frac{\partial^4 w_b}{\partial x^4} + c_b \frac{\partial w_b}{\partial t} = P(x, t) \quad (1)$$

where EI , m_b , and c_b = flexural stiffness, mass per unit length, and damping coefficient of the bridge, respectively; w_b = bridge vertical displacement; and $P(x, t)$ = interacting force between the vehicle and the bridge.

The solution of the motion equation can be obtained by modal superposition. For a simply supported beam, the general equation for the n th mode in the modal coordinate may be written as

$$\ddot{q}_{bn}(t) + 2\xi_{bn}\omega_{bn}\dot{q}_{bn}(t) + \omega_{bn}^2 q_{bn}(t) = \frac{2}{m_b L_b} P_{bn}(t) \quad (2)$$

where $q_{bn}(t)$ = n th modal coordinate, ω_{bn} = natural frequency, ξ_{bn} = damping ratio, and $P_{bn}(t)$ = generalized modal force, which may be expressed in the following form for a series of moving loads (Li et al. 2003):

$$P_{bn}(t) = \sum_{i=K}^M \sin \frac{n\pi(vt - a_i)}{L_b} \left[P_{0,i} - m_s \sum_{k=1}^{\infty} \sin \frac{k\pi(vt - a_i)}{L_b} \ddot{q}_{bk}(t) + c_c \dot{w}_i + k_c w_i \right] \quad (3)$$

For a single vehicle scenario, and neglecting the vehicle mass, the following solution for the displacement of the bridge may be established (Yang and Lin 2005):

$$w_b(x, t) = \sum_n \frac{\Delta}{1 - S_n^2} \left[\sin \frac{n\pi x}{L_b} \left(\sin \frac{n\pi V}{L_b} t - S_n \sin \omega_{bn} t \right) \right] \quad (4)$$

where $\Delta = (-2m_c g L^3)/(n^4 \pi^4 EI)$ = static deflection caused by the passing vehicle with respect to the n th mode, and $S_n = (n\pi V)/(L_b \omega_{bn})$ = normalized speed parameter.

It is clear from Eq. (4) that the dynamic response is affected by the speed, and there exist forced frequency components in the response at an equal interval (circular frequency) of $\pi V/L_b$, i.e.,

$$f_{dr} = \frac{1}{2\pi} \frac{n\pi V}{L_b} = n \frac{V}{2L_b} \quad (5)$$

These are termed driving frequencies (Yang and Lin 2005). Note that L_b/V is the duration of the vehicle passing over the bridge; therefore, they effectively reflect the apparent frequencies associated with the duration of the moving load.

When a series of moving loads are involved, as in the case of a moving train with multiple carriages, the repeated loads may be expressed as

$$P(t) = \sum_j^{N_c} f_c(t) \delta(t - jt_c) \quad (6)$$

where N_c = number of carriages, $f_c(t)$ = generic wheel-axle load, t_c = time interval between the repeated loads, and $t_c = L_c/V$, with L_c = length of a carriage.

Apparently the load described by Eq. (6) is a series (N_c number) of periodic load pulses with a time interval of t_c . Such a load will manifest on the frequency spectrum as having apparent frequencies at $n \times (1/t_c) = nV/L_c$. Despite the fact that, in a real trainload, each carriage involves multiple wheel sets, the frequencies from the trainload are found to be dominated by nV/L_c (Ju et al. 2009). Such apparent frequencies in the trainload are termed dominant frequencies:

$$f_{do,n} = nV/L_c \quad (7)$$

The two types of excitation frequencies mentioned previously essentially involve two normalized speed parameters: $V/2L_b$ and V/L_c . It is generally recognized that when these ratios or their multiples become closer to the bridge natural frequencies, resonant response to a certain degree will occur. These speed ratios have therefore been used as general criteria concerning the bridge resonance (Fryba 2001; Xia et al. 2006). However, there is a lack of specific criteria with regard to the relative importance of different orders (multipliers) of these ratios and how they actually affect the resonance amplitude. Without such specific knowledge, the general resonance criteria will be of little practical use, because a bridge could be exposed to such a resonance condition almost in any speed range, given a choice of the multiplier.

In what follows, the resonance condition will be investigated with an aim to derive quantification of the relative resonance severity among the various possible resonance combinations. First, the critical speed and its variation with the vehicle mass from the perspective of resonance frequencies under a single vehicle load will be examined. The derivation of resonance under a typical trainload with multiple carriages then follows.

Critical Speed: Single Vehicle Scenario

Critical speed is a term used to describe the condition under which a bridge reaches the maximum (global) response when subjected to a single moving load (Fryba 1999; Yang et al. 2004). Without considering the moving mass, the critical speed can be established mathematically (Fryba 1999; Yang et al. 2004) as

$$V_{cr} = 2f_1 L \quad (8)$$

where f_1 = bridge fundamental frequency, and L = bridge length.

Recalling the definition of the apparent driving frequencies in Eq. (5), it can be seen that the critical speed is essentially one that results in the first driving frequency ($n = 1$) being equal to the bridge fundamental frequency f_1 . From this point of view, the critical

speed condition may be regarded as the bridge resonance under a single moving load.

When the mass of the vehicle is also taken into account, the critical speed may be identified by tracing the bridge response under different speeds. This may be done by solving the beam motion equation involving the vehicle mass as follows (Dehestani et al. 2009):

$$m_b \frac{\partial^2 w_b}{\partial t^2} + EI \frac{\partial^4 w_b}{\partial x^4} + M \left(V^2 \frac{\partial^2 w_b}{\partial x^2} + 2V \frac{\partial^2 w_b}{\partial t \partial x} + \frac{\partial^2 w_b}{\partial t^2} \right) \times \delta(x - Vt) = -Mg\delta(x - Vt) \quad (9)$$

where the third term represents the inertial effect caused by the moving mass M . The rest of the notation follows the definition given in the previous section.

Using Eq. (9), the bridge deflection under a single moving vehicle at different speeds may be calculated using numerical techniques, such as the Runge-Kutta method. The critical speed can then be identified when bridge deflection reaches the maximum value. Results show the critical speed becomes smaller when the moving mass is involved (Esmailzadeh and Ghorashi 1995; Dehestani et al. 2009).

The change of the critical speed caused by the vehicle mass may be deduced by the change (decrease) of the resonance frequency of the responding system as a result of the involvement of the moving mass:

$$V_{cr,m} = 2f_m L \quad (10)$$

where $V_{cr,m}$ = critical speed considering the effect of moving mass, and f_m = effective natural frequency of the bridge during the passage of the moving vehicle.

The calculation of the critical speed then becomes a matter of finding such an effective (or equivalent) natural frequency of the bridge-vehicle system. For simply supported or fixed ends, the bridge-vehicle system will attain the lowest fundamental frequency, f_{m1} , when the vehicle approaches the midspan of the bridge. The effective natural frequency, f_m , should lie between f_{m1} and f_1 . Introducing an effective frequency ratio α , such that $f_m = \alpha f_{m1}$, Eq. (10) can be rewritten as

$$V_{cr,m} = 2\alpha f_{m1} L \quad (11)$$

where α is within $(1, f_1/f_{m1})$.

In the current study, numerical investigations are carried out using the finite-element (FE) model to examine the variation of the critical speed with the amount of the moving mass and hence the factor α . The general FE model is developed for the analysis of a generic beam-bridge subjected to a series of moving loads representing a moving train, as schematically illustrated in Fig. 1(a), where the moving vehicles are represented by loaded mass blocks while spring elements may be added to represent the vehicle suspension system. For simplicity and without affecting the main frequency characteristics, herein the mass blocks are attached directly on the bridge surface using a surface-to-surface contact approach, as shown in Fig. 1(b).

For a single vehicle scenario, only one mass block is needed. The bridge is assigned an arbitrary length of 8 m and a mass of 16,000 kg, whereas the stiffness is specified such that the bare bridge has a fundamental natural frequency of 14 Hz. For such a bridge, the critical speed for a moving vehicle without considering the vehicle mass can be calculated according to Eq. (8): $V_{cr} = 2f_1 L = 2 \times 14 \times 8 = 224$ m/s.

Using the FE model, different amounts of the vehicle mass, in terms of the vehicle-to-bridge mass ratio, are simulated. For each amount of the mass ratio, the response of the bridge is analyzed under different moving speeds, and the critical speed, $V_{cr,m}$, for which the bridge attains the maximum (midspan) response, is identified. Besides, for each mass ratio, f_{m1} can be calculated using the FE model by placing the mass at the midspan. Substituting these parameters into Eq. (11), the corresponding effective frequency ratio α can be established. Fig. 2 shows the variation of α with the vehicle-to-bridge mass ratio.

From Fig. 2 it can be found that when the vehicle-to-bridge mass ratio is small, in particular smaller than 1.0 in the short span bridge example herein, the value of α may be approximately taken as equal to unity, which means the lower-bound natural frequency with the moving mass, f_{m1} , can be used directly as the effective bridge frequency to obtain the critical speed. When the vehicle-to-bridge mass ratio is large than 1.0, however, α tends to become noticeably larger than unity, indicating that the effective frequency would be

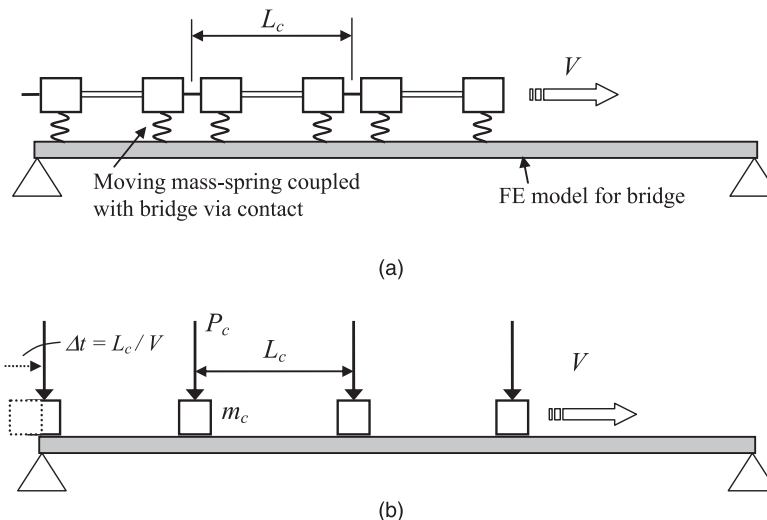


Fig. 1. Schematic of a general FE model for bridge dynamic response under a moving train: (a) general FE model configuration; (b) simplified FE model with equally spaced moving mass/load via surface contact (only one mass block is needed for a single vehicle scenario)

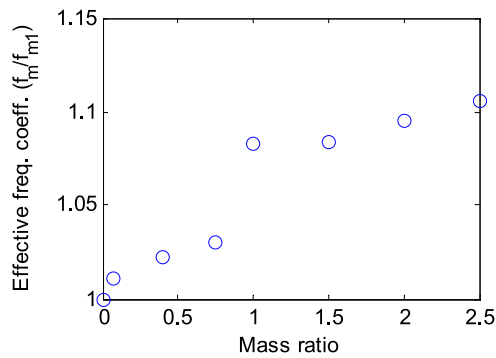


Fig. 2. Correlation between vehicle-bridge mass ratio and effective frequency ratio α

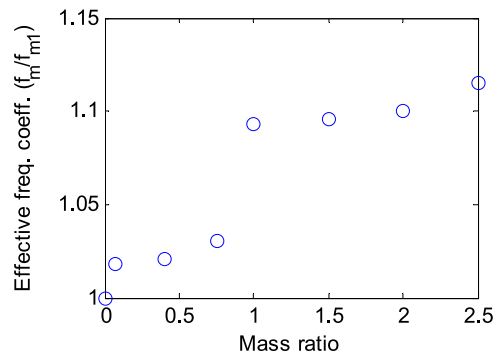


Fig. 3. Correlation between vehicle-bridge mass ratio and frequency ratio α for a bridge with fixed-fixed ends

markedly higher than the lower-bound frequency, and this should be considered in the calculation of the critical speed. For the mass ratio ranging from 1.0 to 2.5, α is approximately equal to 1.1.

Extending from the preceding analysis for a simply supported condition, the bridge with both ends fixed is also analyzed to examine the moving mass effect on the critical speed. The fundamental natural frequency is about 31 Hz, so according to Eq. (8) the theoretical critical speed would be about 500 m/s without considering the moving mass. Fig. 3 depicts the correlation between vehicle-bridge mass ratio and the effective frequency ratio α . Similar observations as for the simply supported case can be observed, i.e., when the mass ratio is smaller than 1.0, α is close to unity, and when the mass ratio is larger than 1.0, α increases to about 1.1.

Resonance and Resonance Severity under Trainload with Multi-carriages: Formulation

When multiple carriages are involved, the frequency of the excitation will feature the apparent dominant frequencies that associate with time intervals between consecutive loads, in addition to the driving frequencies discussed in the previous section. Consequently, the resonance is expected to exhibit a different character beyond the context of the critical speed. It has been generally understood (Ju and Lin 2003; Xia et al. 2006) that when a specific dominant frequency from the multiple loads coincides with one of the first few bridge natural frequencies, especially the first bridge natural frequency, resonance to a certain degree would occur.

Li and Su (1999) studied the amplification-suppression phenomenon between consecutive moving loads when a series of such

loads moves on the bridge at certain resonant speeds based on the principle of superposition. Assuming the bridge response is governed by the fundamental mode, the dynamic response can be expressed using a generalized coordinate as

$$w_b(x, t) = q(t)\phi(x) \quad (12)$$

where $q(t)$ = generalized coordinate that expresses the bridge vertical response with time, and $\phi(x) = \sin(\pi x/L_b)$ = bridge fundamental mode shape. Thus, the equation of motion of the bridge under multiple moving loads may be written in a generalized form as

$$\ddot{q}(t) + 2\xi\omega\dot{q}(t) + \omega^2q(t) = \frac{2}{mL_b} \sum_{i=K}^{M+K} \sin \frac{\pi(vt - a_i)}{L_b} P_i \quad (13)$$

where M = number of carriages currently acting on the bridge. The solution of $q(t)$ may be sought by first examining the forced vibration of the bridge under a single moving load and the free vibration after the passage of the single load and subsequently using the principle of superposition. For the convenience of the superposition operation, consider the single moving load as the i th load in the moving load series, and let $a_i = (i - 1)L_c$ be the distance of the i th load from the first load, and $t = 0$ be the time when the first moving load enters the bridge. Thus, the forced vibration caused by the i th moving load takes place when $(a_i/v) \leq t \leq (a_i/v + L_b/v)$. Specializing Eq. (13) for the i th single moving load only, the solution can be expressed as the combination of a particular (forced or steady-state) term and a homogeneous (free or transient) term as

$$q(t) = \frac{y_{st}}{1 - \beta^2} \sin \frac{\pi(vt - a_i)}{L_b} - y_{st} \frac{\beta}{1 - \beta^2} e^{-\xi\omega(t - a_i/v)} \sin \omega \left(t - \frac{a_i}{v} \right) \quad (14)$$

where y_{st} = static midspan deflection with moving load P acting on the midspan, and $y_{st} = PL_b^3/48EI$ in a simply supported case. $\beta = \pi v/L_b\omega$, with ω being the bridge fundamental (circular) frequency; it should be noted that when the carriage mass is considered, ω is understood to be the mass-loaded bridge frequency. ξ is the damping ratio.

The free vibration after the passage of the single (i th) moving load, i.e., $t \geq t' = a_i/v + L_b/v$, is determined by the displacement $q_i(t')$ and velocity $\dot{q}_i(t')$ at the end of forced vibration period t'

$$q(t) = e^{-\xi\omega(t - t')} \left[\frac{\dot{q}_i(t') + \xi\omega q_i(t')}{\omega} \sin \omega(t - t') + q_i(t') \cos \omega(t - t') \right] \quad (15)$$

Substituting $q_i(t')$ and $\dot{q}_i(t')$ from Eq. (14) into Eq. (15) yields

$$q(t) = -e^{-\xi\omega(t - a_i/v - L_b/v)} \frac{y_{st}\beta}{1 - \beta^2} \sin \omega \left(t - \frac{a_i}{v} - \frac{L_b}{v} \right) - e^{-\xi\omega(t - a_i/v)} \frac{y_{st}\beta}{1 - \beta^2} \sin \omega \left(t - \frac{a_i}{v} \right) \quad (16)$$

With Eqs. (14) and (16), the bridge response under a series of moving loads can be established using superposition. Consider a generic

scenario of $(K - 1)$, for which $(K - 1)$ moving loads have crossed the bridge, while M moving loads are acting on the bridge; M is determined by the bridge-to-carriage length ratio: $M = \text{Int.}(L_b/L_c)$. It follows that $(a_{K-1}/v + L_b/v) \leq t \leq (a_K/v + L_b/v)$, and the bridge response can be expressed in accordance with the solution by Li and Su (1999) as

$$q(t) = \frac{y_{st}}{1 - \beta^2} \times \sum_{i=K}^{M+K-1} \sin \frac{\pi(vt - a_i)}{L_b} - \frac{y_{st}\beta}{1 - \beta^2} \times \sum_{i=K}^{M+K-1} e^{-\xi\omega(t-a_i/v)} \sin \omega \left(t - \frac{a_i}{v} \right) - \frac{y_{st}\beta}{1 - \beta^2} \times \sum_{i=1}^{K-1} e^{-\xi\omega(t-a_i/v-L_b/v)} \sin \omega \left(t - \frac{a_i}{v} - \frac{L_b}{v} \right) - \frac{y_{st}\beta}{1 - \beta^2} \times \sum_{i=1}^{K-1} e^{-\xi\omega(t-a_i/v)} \sin \omega \left(t - \frac{a_i}{v} \right) \quad (17)$$

In the preceding equation, the first two terms represent the bridge vibration caused by the acting M moving loads, whereas the last two terms are the bridge free vibration caused by the preceding $(K - 1)$ moving loads. Note that the second and fourth terms can be combined to yield a summation from $i = 1$ to $(M + K - 1)$. It is clear that the first term in Eq. (17), i.e., the steady-state term of the bridge forced vibration, does not increase as the moving loads pass over the bridge. Thus, the bridge resonance can be analyzed by only investigating the remaining terms, denoted as $q_r(t)$, in Eq. (17). As stated earlier, the prior condition for the resonance to occur is that one of the dominant frequencies from multiple moving loads, expressed in Eq. (7), equals the bridge fundamental frequency, i.e.,

$$n \frac{v_{re}}{L_c} = f_1 = \frac{\omega}{2\pi} \quad (18)$$

where v_{re} = resonant speed herein. Rearrange the preceding equation,

$$v_{re} = \frac{f_1 L_c}{n} = \frac{\omega L_c}{2\pi n} \quad (19)$$

Substituting the preceding specific resonant speed and $a_i = (i - 1)L_c$ into Eq. (17), the last three terms can be rewritten as

$$q_r(t) = \left(-y_{st} \frac{\beta_{re}}{1 - \beta_{re}^2} e^{-\xi\omega t} \right) \times \left[\left(\sum_{i=1}^{M+K-1} e^{\xi\omega a_i/v_{re}} \right) \times \sin \omega t + \left(\sum_{i=1}^{K-1} e^{\xi\omega(a_i/v_{re} + L_b/v_{re})} \right) \times \sin \omega \left(t - \frac{L_b}{v_{re}} \right) \right] \quad (20)$$

where

$$\beta_{re} = \frac{\pi v_{re}}{L_b \omega} \quad (21)$$

or

$$q_r(t) = \left(-y_{st} \frac{\beta_{re}}{1 - \beta_{re}^2} e^{-\xi\omega t} \right) \times \left[\left(\sum_{i=1}^{M+K-1} e^{2\pi n(i-1)\xi} \right) \times \sin \omega t + \left(\sum_{i=1}^{K-1} e^{2\pi n[(i-1)+L_b/L_c]\xi} \right) \times \sin \left(\omega t - 2\pi n \frac{L_b}{L_c} \right) \right] \quad (22)$$

where

$$\beta_{re} = \frac{L_c}{2nL_b} \quad (23)$$

It can also be deduced that under the resonant speed of Eq. (19), the time interval between two consecutive moving loads is $L_c/v_{re} = 2\pi n/\omega$; this makes the sine functions in Eq. (20) repeat themselves while their coefficients cumulate as the consecutive moving loads pass over the bridge. As noted by Li and Su (1999), this means the response of the bridge under a series of moving loads at the resonance speed will always tend to increase. It is the rate of increase between the consecutive moving loads that determines the severity of the resonance effect.

From Eq. (20), it can be readily shown that when $\cos(\omega L_b/v_{re}) = 1$, the two sine terms get in phase, and when $\cos(\omega L_b/v_{re}) = -1$, they are out of phase. Thus, as Li and Su (1999) suggest, $\cos(\omega L_b/v_{re}) = 1$ may be used to identify the most serious bridge resonance, whereas $\cos(\omega L_b/v_{re}) = -1$ sets out the condition by which the most significant suppression effect (minimum bridge response) occurs.

Recalling Eq. (19), it follows that $\omega L_b/v_{re} = 2\pi n L_b/L_c$; thus, it becomes clear that only at special combinations of the bridge and carriage lengths ($L_b/L_c = \text{integer or at } 0.5 \text{ interval}$) will the maximum resonance or suppression as mentioned previously take place. For all other and more general length combinations, the value of $\cos(\omega L_b/v_{re}) = \cos(2\pi n L_b/L_c)$ could be anything between -1 and 1 ; therefore, in such cases, the severity of the resonance effect cannot be determined using the preceding condition.

For simplicity and without losing generality, herein damping is ignored; thus, Eq. (20) can be rewritten as

$$q_r(t) = \left(-y_{st} \frac{\beta_{re}}{1 - \beta_{re}^2} \right) \times \left[(M + K - 1) \times \sin \omega t + (K - 1) \times \sin \omega \left(t - \frac{L_b}{v_{re}} \right) \right] \quad (24)$$

Hence,

$$q_r(t) = A_{k-1} \sin(\omega t - \theta_{k-1}) \quad \text{with} \quad (a_{K-1} + L_b)/v \leq t \leq (a_K + L_b)/v \quad (25)$$

where A_{k-1} = bridge response amplitude for the $(K - 1)$ th scenario when $(K - 1)$ moving loads have left the bridge, whereas M moving loads act on the bridge

$$A_{k-1} = \frac{y_{st}\beta_{re}}{1 - \beta_{re}^2} \sqrt{\left[(M + K - 1) + (K - 1)\cos\left(\omega \frac{L_b}{v_{re}}\right) \right]^2 + \left[(K - 1)\sin\left(\omega \frac{L_b}{v_{re}}\right) \right]^2} \quad (26)$$

and

$$\tan(\theta_{k-1}) = \frac{(K-1)\sin\left(\omega \frac{L_b}{v_{re}}\right)}{(M+K-1) + (K-1)\cos\left(\omega \frac{L_b}{v_{re}}\right)}$$

Now consider the immediate next (K th) scenario, i.e., next moving load enters the bridge, or in other words, K moving loads have left the bridge: $q_r(t)$ becomes

$$q_r(t) = A_k \sin(\omega t - \theta_k) \quad \text{with} \quad (a_K + L_b)/v \leq t \leq (a_{K+1} + L_b)/v \quad (27)$$

where

$$A_k = \frac{y_{st}\beta_{re}}{1-\beta_{re}^2} \sqrt{\left[(M+K) + K\cos\left(\omega \frac{L_b}{v_{re}}\right)\right]^2 + \left[K\sin\left(\omega \frac{L_b}{v_{re}}\right)\right]^2} \quad (28)$$

By comparing amplitudes of bridge responses using Eqs. (26) and (28), the change of bridge response under consecutive moving loads can be evaluated. To simplify the calculation, without affecting the final conclusion, the squared amplitude is used. Thus, the amplitude (squared) difference in the response between consecutive moving loads can be expressed as

$$\begin{aligned} (A_K)^2 - (A_{K-1})^2 &= \left(\frac{y_{st}\beta_{re}}{1-\beta_{re}^2}\right)^2 \times \left\{ \left[(M+K) + K\cos\frac{\omega L_b}{v_{re}}\right]^2 + \left(K\sin\frac{\omega L_b}{v_{re}}\right)^2 \right. \\ &\quad \left. - \left[(M+K-1) + (K-1)\cos\frac{\omega L_b}{v_{re}}\right]^2 - \left[(K-1)\sin\frac{\omega L_b}{v_{re}}\right]^2 \right\} \\ &= \left(\frac{y_{st}\beta_{re}}{1-\beta_{re}^2}\right)^2 \times 2(M+2K-1) \left(1 + \cos\frac{\omega L_b}{v_{re}}\right) \end{aligned} \quad (29)$$

Let

$$Z = \left(\frac{\beta_{re}}{1-\beta_{re}^2}\right)^2 \times \left(1 + \cos\frac{\omega L_b}{v_{re}}\right) = Z_a \times Z_b \quad (30)$$

It follows

$$\frac{(A_K)^2 - (A_{K-1})^2}{y_{st}^2} = Z[2(M+2K-1)] \quad (31)$$

The coefficient Z , along with its components Z_a and Z_b as expressed in Eq. (30), represents the rate of increase of the bridge response amplitudes as the moving loads pass over the bridge at a potential resonance speed. Therefore, it can serve as a direct indicator of the severity of the resonance at such a speed. From Eq. (30), it can also be seen that Z is a nonnegative coefficient; this further confirms the observation mentioned earlier that the bridge response will tend to always increase under a series of moving loads at a resonant speed.

The preceding derivations are valid under the presumption that the moving loads travel at a resonance speed, i.e., Eq. (19) holds. Substituting Eqs. (19) and (23) into Eq. (30) yields

$$\begin{aligned} Z &= Z_a \times Z_b = \left[\frac{L_c/2nL_b}{1-(L_c/2nL_b)^2}\right]^2 \times \left[1 + \cos\left(2\pi n \frac{L_b}{L_c}\right)\right] \\ &= \left[\frac{2nL_b/L_c}{(2nL_b/L_c)^2 - 1}\right]^2 \times \left[1 + \cos\left(2\pi n \frac{L_b}{L_c}\right)\right] \end{aligned} \quad (32)$$

Eq. (32) is significant because it shows that, when the resonance condition is met, i.e., $v = v_{re} = \omega L_c/2\pi n$ (or $f_1 L_c/n$), the relative severity of the actual resonance is entirely determined by the bridge-to-carriage length ratio L_b/L_c . Because for any given bridge-train system, L_b/L_c is a constant, the Z value can be obtained for different orders of resonant speeds (with different n values) as a function of L_b/L_c . Fig. 4 depicts the relationship between Z , Z_a , Z_b , and L_b/L_c .

The following can be observed from Fig. 4:

1. While Z_b oscillates in the range of (0, 2) as the ratio L_b/L_c varies, Z_a is localized around the singularities at $L_b/L_c = 1/2n$, with the largest singular ratio at 0.5.
2. As a result, the Z value is dominated by Z_a and tends to concentrate in the lower L_b/L_c ratio range up to about 1.5. For longer bridges with $L_b/L_c > 1.5$, the resonance effect caused by the excitation of multiple moving loads, or essentially the dominant frequencies associated with the repeated loads, appear to diminish.
3. A significant resonance effect to be excited by a higher-order dominant frequency, i.e., $n \geq 2$, would be possible only for very short bridges with $L_b/L_c \leq 0.7$; and in this range, especially when $L_b/L_c \leq 0.5$, the bridge could be excited into a certain level of resonance at more than one occasion of n , which implies that there exists more than one critical resonance speed. Note that the critical speed is inversely related to n ; thus, real resonance at lower speeds becomes possible in such cases. On the other hand, for $L_b/L_c > 0.7$ (and less than 1.5), a significant resonance effect is only associated with the first-order dominant frequency, i.e., $n = 1$.

To give an example, consider a standard carriage length $L_c = 20$ m and two cases of bridges:

1. $L_b = 8$ m and $f_1 = 14$ Hz; thus, $L_b/L_c = 0.4$. From Fig. 4, significant Z could arise from three different n values: $n = 1$, $Z = 0.95$; $n = 2$, $Z = 1.37$; and $n = 3$, $Z = 0.33$. From Eq. (19), the corresponding resonance speeds can be found as $v_{re} = f_1 L_c/n = 280$, 140, and 93.3 m/s (or 336 km/h), respectively.
2. $L_b = 18$ m and $f_1 = 7$ Hz; thus, $L_b/L_c = 0.9$. From Fig. 4, significant Z could only arise from $n = 1$, resulting in $Z = 1.17$. The corresponding resonance speed is: $v_{re} = f_1 L_c/n = 140$ m/s. Less significant but still appreciable resonance may be excited at $n = 2$, with $Z = 0.12$, but it is noteworthy that the corresponding speed is reduced to a more attainable level of 70 m/s or 250 km/h.

Compared with the use of $\cos(\omega L_b/v_{re})$ suggested in Li and Su (1999), the Z factor developed previously is a much more robust and stable indicator of the resonance effect associated with the excitation of multiple moving loads at a potential resonance speed. Further examination in this regard will be illustrated in the next section.

It should be pointed out that the current discussion is restricted to the situation whereby the dynamic response is governed by the fundamental mode of the bridge, and the resonance effect is also confined to the resonance at the fundamental mode frequency, i.e., when one of the dominant frequencies from the multiple moving load excitation coincides with the fundamental mode

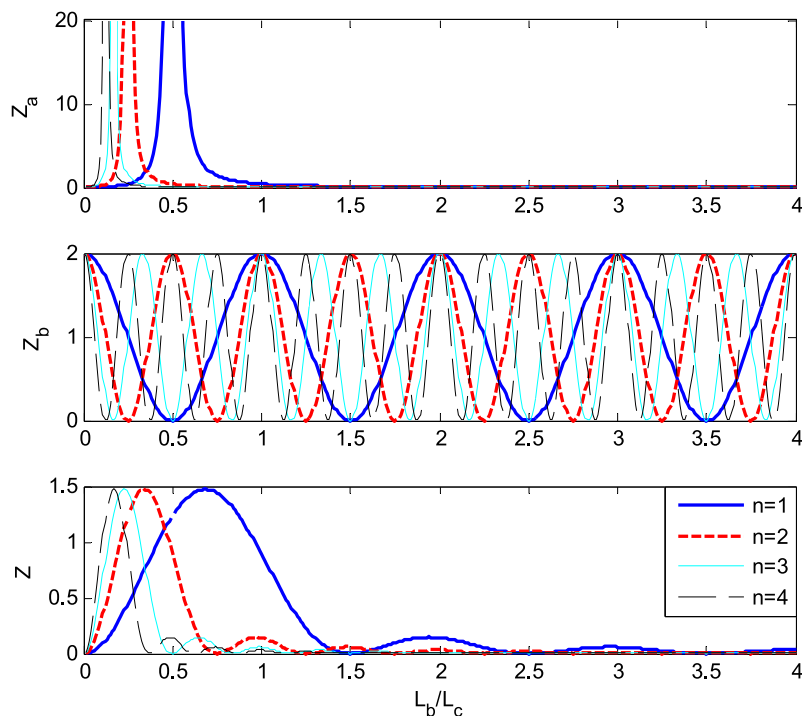


Fig. 4. Z_a , Z_b , and Z as functions of L_b/L_c

frequency of the bridge. Therefore, the observations may be considered as applicable for small- to medium-length bridges.

When the carriage mass is involved, the bridge fundamental frequency f_1 will be altered. If the alteration is not significant or the change may be considered as constant (this could happen only with relatively long bridges that may be considered as being constantly loaded with a given number of carriage masses), the discussions presented in this section are still generally applicable, except that the natural frequency of the bridge should be replaced with the mass-loaded frequency. Conversely, if the variation of the effective mass of the system is significant and the frequency of the system cannot be treated as a constant, strictly speaking, there will be no clearly defined resonance speeds ($v_{re} = f_1 L_c / n$). In such cases, similar to the situation discussed in the earlier section for the critical speed under a single moving vehicle, an effective natural frequency of the bridge-moving mass system may be used to determine the resonance speeds under multiple carriages. The corresponding resonance severity can then be established using the Z factor expressed in Eq. (32). This will be demonstrated in the numerical examples in the section that follows, along with general verification of the effectiveness of the Z factor where the moving mass is negligible.

Resonance and Resonance Severity under Trainload with Multi-carriages: Numerical Verification

Two illustrative sets of examples, including a short-span and a medium-span bridge, are provided to demonstrate the effectiveness of Z factor in Eqs. (30) and (32) in the assessment of the bridge resonance under the excitation of multiple moving loads, without the involvement of the carriage mass. The numerical model is similar to that described earlier in the evaluation of critical speed (Fig. 1). To simulate multiple carriages, a series of moving blocks are set to start moving consecutively from the left side of the bridge, at a time interval of L_c/v .

Table 1. Z and $\cos(\omega L_b/v_{re})$ Values at Potential Resonant Speeds for a 32-m Bridge without Considering Carriage Mass

Parameters	n				
	1	2	3	4	5
$v_{re}(\text{m/s})$	111.3	55.65	37.1	27.83	22.26
Z	0.0015	0.0556	0.0012	0.0125	0.0012
$\cos(\omega L_b/v_{re})$	-0.99	0.96	-0.9	0.83	-0.73

Moving Mass Negligible

The properties of a bridge-train system as used in Li and Su (1999) are first used to allow for a direct comparison. The bridge is 32 m long, with a fundamental frequency at 5.3 Hz, and the train is modeled using a series of 20 moving loads with an interval distance of 21 m. Based on these parameters, the resonant speeds v_{re} can be calculated using Eq. (19), and the corresponding Z can be obtained using Eq. (30). For an observation, the values of $\cos(\omega L_b/v_{re})$, as used by Li and Su (1999) for identifying the maximum and minimum resonance effects, are also calculated. The results are summarized in Table 1.

From Table 1, Z reaches the maximum value at $n = 2$, and the corresponding resonance speed is 55.65 m/s. This indicates that the most severe bridge resonance will occur under the preceding speed, and it is the second dominant excitation frequency that causes the resonance. For this particular case, the prediction by means of $\cos(\omega L_b/v_{re})$ is consistent with the current result. However, as discussed earlier in association with Fig. 4, the criterion of $\cos(\omega L_b/v_{re})$ is suited for the detection of the maximum and minimum resonance effect where the cosine value approaches 1.0 or -1.0, which happens to be the case in this example. Otherwise, as will be shown in the 8-m bridge case later, this cosine criterion will become meaningless.

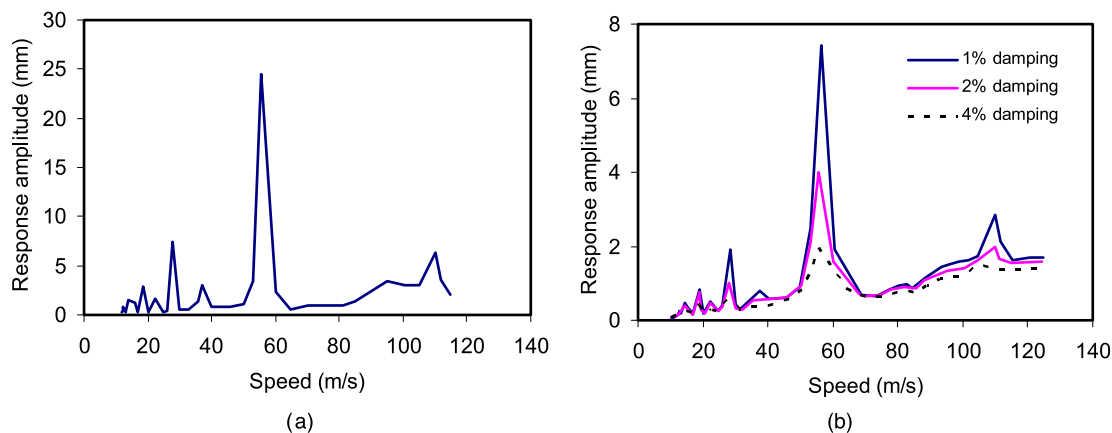


Fig. 5. Variation of midspan deflection with moving speed for a 32-m bridge without considering carriage mass (after Li and Su 1999): (a) without damping; (b) with damping

Fig. 5 shows the variation of the midspan deflection of the 32-m bridge with the moving speed without considering carriage mass. From Fig. 5(a), where the damping is ignored, it can be found clearly that the most serious resonance is excited at about 55 m/s, whereas the next peak appears at about 28 m/s. These results agree perfectly with the predictions in accordance with the Z values mentioned previously. Results for lightly damped systems with a damping ratio being 1, 2, and 4%, respectively [Fig. 5(b)], appear to exhibit the same order of resonances at the respective resonance speeds as in the undamped system, while the response amplitudes reduce as the damping increases, as can be expected. This indicates that the Z factor is equally applicable in lightly damped systems.

A bridge of 8 m length, as used earlier in the analysis of the critical speed under a single moving load, is then used to demonstrate the effectiveness of Z in Eq. (30) in a short-span bridge scenario. A train model including 10 carriages with a length equal to 20 m each is considered. The bridge is analyzed here without considering the carriage mass, whereas the carriage load on the bridge is retained at a normal carriage load level of 400 kN. Both undamped and damped (1%) analyses are carried out.

A prediction of the resonant speeds and the corresponding Z values is made using Eqs. (19) and (30), similar to the previous medium-span bridge case. Table 2 lists the prediction results, along with the values using the criterion of $\cos(\omega L_b/v_{re})$.

From Table 2, Z reaches the maximum value at $n = 2$, and the corresponding resonance speed is 140 m/s. This means the second dominant frequency can cause the most severe bridge resonance. In contrast, the values of $\cos(\omega L_b/v_{re})$ tend to give a rather different picture, with the highest value corresponding to $n = 5$, i.e., at 56 m/s resonance speed.

Fig. 6 shows the variation of the midspan deflection of the bridge with train speed based on the FE analysis results. It can be seen clearly that, without or with damping, the most serious bridge resonance occurs at around 140 m/s, followed by the second highest peak at around 280 m/s, and then the third at around 100 m/s. These agree very well with the prediction from the Z factor, whereas the values of $\cos(\omega L_b/v_{re})$ become meaningless in this case.

Moving Mass Significant

As mentioned in the Resonance and Resonance Severity under Trainload with Multi-carriages: Formulation section, when the moving carriage mass becomes significant, a modified bridge resonance frequency, i.e., the effective fundamental frequency of the bridge-moving

Table 2. Z and $\cos(\omega L_b/v_{re})$ Values at Potential Resonant Speeds for an 8-m Bridge without Considering Carriage Mass

Parameters	n				
	1	2	3	4	5
v_{re} (m/s)	280	140	93	70	56
Z	0.94	1.37	0.33	0.02	0.14
$\cos(\omega L_b/v_{re})$	-0.81	0.31	0.31	-0.81	1

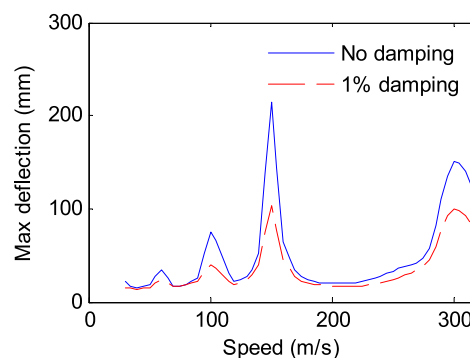


Fig. 6. Variation of midspan deflection with moving speed for an 8-m bridge without considering carriage mass

mass system, needs to be established before the Z factor may be applied for the determination of the resonance effect. In what follows, two bridge cases with two different levels of the carriage-to-bridge mass ratio are analyzed using the FE model, and the results are examined.

The first case is the same 8-m short-span bridge analyzed in the previous section, with, however, a moving mass of 40,000 kg, which represents the mass of a single laden carriage in a typical high-speed train. This gives rise to a carriage-to-bridge mass ratio of about 2.5. Compared with a fundamental frequency of 14 Hz for the bare bridge, the coupled bridge-moving mass system now has a natural frequency varying from 14 to 6 Hz when the mass moves to the midspan. Clearly, with such large variation of the natural frequency, calculating the resonance speeds using Eq. (19), i.e., $v_{re} = f_1 L_c / n$, with the bare bridge natural frequency would not make sense.

From the evaluation presented in the section Critical Speed: Single Vehicle Scenario, the effective natural frequency under the present moving mass ratio would be around 1.1 times the lowest system frequency, i.e., $1.1 \times 6 = 6.6$ Hz. The corresponding resonance speeds, $v_{re,m} = f_{1,m}L_c/n$, and the associated Z values are shown in Table 3. For a comparison, the resonance speeds as calculated from the bare bridge frequency of 14 Hz are also listed.

The results tend to indicate that the highest resonance effect (highest Z) will take place at a resonance speed of around 64 m/s, with its second dominant frequency coinciding with the effective natural frequency of the bridge-carriage system, and the second highest resonance will be at 132 m/s with its first dominant frequency, followed by 43 m/s with its third dominant frequency.

For verification, the computed peak midspan deflections for the bridge-moving mass system under different moving speeds, using the FE model, are illustrated in Fig. 7. It can be immediately observed that the highest resonance peak indeed takes place at around 60 m/s. The second and third resonance peaks appear to occur at

Table 3. Z and $\cos(\omega L_b/v_{re})$ Values at Potential Resonant Speeds for an 8-m Bridge Considering Carriage Mass

Parameters	n				
	1	2	3	4	5
v_{re} (m/s)	280	140	93	70	56
$v_{re,m}$ (m/s)	132	64	43	34	26
Z	0.94	1.37	0.33	0.02	0.14
$\cos(\omega L_b/v_{re})$	-0.81	0.31	0.31	-0.81	1

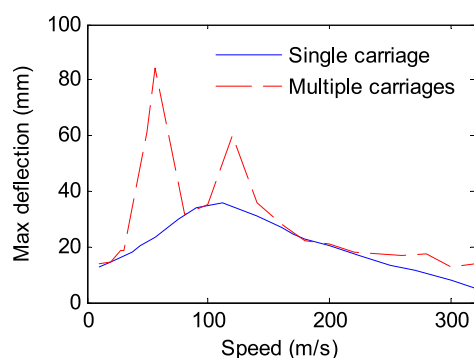


Fig. 7. Variation of maximum midspan deflection with moving speed for an 8-m bridge considering carriage mass

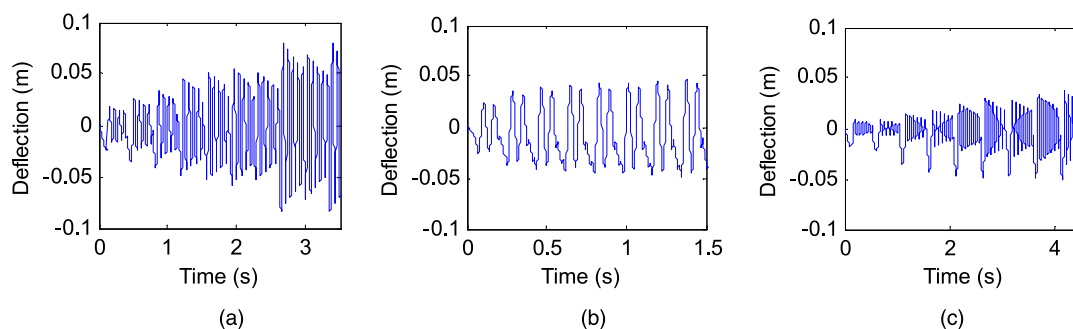


Fig. 8. Midspan deflection time histories for the 8-m bridge under various resonance speeds: (a) 60 m/s; (b) 120 m/s; (c) 40 m/s

around 120 and 40 m/s, respectively. These results are in good agreement with the predictions using the Z factor in conjunction with the effective natural frequency of the bridge-mass system. It is worth noting that the calculations based on the bare bridge frequency fail to predict the actual resonance responses, as expected.

Shown in Fig. 7 is also the peak response curve when the bridge is subjected to a single carriage of the same load and mass as used in the multiple-carriage analysis. It can be observed that in the single carriage situation the peak response occurs at a different speed from the resonance in the multiple-carriage scenario. The critical speed for the single carriage load herein is found to be around 105 m/s, and this is consistent with the prediction using Eq. (10): $V_{cr,m} = 2\beta f_{m0}L_B = 2 \times 1.1 \times 6 \times 8 = 105.6$ m/s. Understandably, the maximum resonance effect under a single carriage load is considerably lower than its counterpart under multiple-carriage loads.

Fig. 8 presents the midspan deflection time histories when the bridge is subjected to the series of moving carriages at around the three resonance speeds. The severity of the resonance effect under the speed of 60 m/s is remarkable, causing a maximum deflection of 0.07 m. The resonance effects at 120 and 40 m/s are also noticeable, resulting in a maximum deflection of about 0.05 and 0.03 m, respectively. These trends are consistent with the predictions using the Z factor as mentioned earlier.

The next example case is a 40-m-long bridge with a fundamental frequency (bare bridge) of around 10 Hz. The mass of the bridge is assumed to be 100,000 kg, considering its length. The same train model as used in the previous 8-m example is used, i.e., 10 carriages and each having a mass of 40,000 kg. The carriage-to-bridge mass ratio is thus 40%. The range of variation of the natural frequency of the bridge system considering the moving mass is found to be 9–10 Hz.

According to Fig. 4, for the present bridge with $L_b/L_c = 2.0$, the most significant resonance effect can be expected to associate with the first-order dominant frequency, i.e., $n = 1$, and the corresponding Z value is 0.14. For the determination of the resonance speed, an effective natural frequency of the mass loaded bridge is estimated to be approximately 9.3 Hz. The resonance speeds can then be calculated using this effective bridge frequency. Because the range of variation of the natural frequency is not large in this case, these critical speeds do not deviate significantly from those calculated according to the natural frequency of the bare bridge. Table 4 summarizes the two sets of the critical speeds with n from 1 to 5 and the corresponding Z and $\cos(\omega L_b/v_{re})$ values.

From Table 4, for the particular case under consideration, $\cos(\omega L_b/v_{re})$ yields a value of 1.0 for all five resonant speeds, which does not provide a useful indication of the relative resonance effects at these different speeds. In contrast, the Z values indicate clearly that

the highest resonance occurs when $n = 1$, while the moving speed is about 186 (using the effective natural frequency) or 200 m/s (using bare bridge natural frequency).

The results from the FE model analysis are shown in Fig. 9 in terms of the variation of midspan deflection with moving speed, without or with consideration of (1%) damping. It can be found clearly that the most severe resonance does occur at around 186 m/s, followed by resonances at around 90 and 60 m/s. Similar to the short bridge example, these results agree very well with the predictions using the Z factor.

Fig. 10 shows the computed bridge midspan deflection time histories with 1% damping for moving speeds of 200, 190, and 100 m/s, respectively. It can be seen clearly that much more serious resonance is excited at 190 m/s compared with that at 100 m/s.

Conclusions

The resonance of railway bridges under trainload excitation is investigated in light of the frequency characteristics of the excitation,

Table 4. Z and $\cos(\omega L_b/v_{re})$ Values at Resonant Speeds for a 40-m Bridge Considering Carriage Mass

Parameters	n				
	1	2	3	4	5
v_{re} (m/s)	200	100	67	50	40
$v_{re,m}$ (m/s)	186	93	62	46.5	37.2
Z	0.14	0.03	0.01	0.008	0.005
$\cos(\omega L_b/v_{re})$	1	1	1	1	1

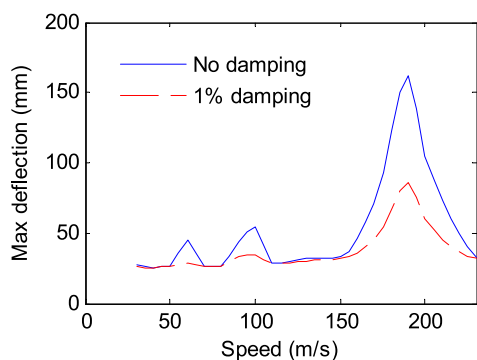


Fig. 9. Variation of maximum midspan deflection with moving speed for a 40-m bridge considering carriage mass

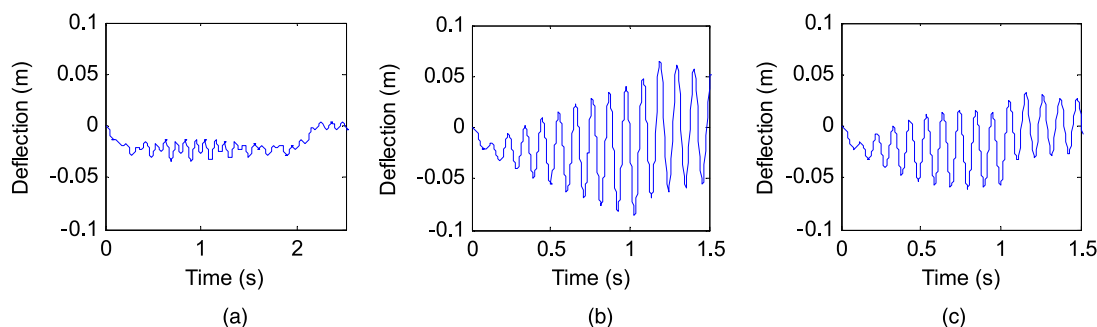


Fig. 10. Midspan deflection time histories of the 40-m bridge under various moving speeds: (a) 100 m/s; (b) 190 m/s; (c) 200 m/s

as well as the varying resonance frequency of the bridge-vehicle system when the vehicle mass is involved. On the basis of the results, the following conclusions may be drawn:

1. For the basic scenario where only a single carriage (represented by a single mass and load) is involved, the variation of the dynamic effect with speed may be interpreted by means of the driving frequency of the excitation, which is an apparent frequency associated with the duration of the load moving over the bridge, i.e., $V/2L_b$. The classical definition of the critical speed then becomes simply a resonant condition such that the driving frequency matches the bridge natural frequency. It is clear that the critical speed reduces as the amount of moving mass increases. The modified critical speed can still be well predicted using the preceding resonance condition, with, however, the consideration of an effective bridge frequency instead of the bare bridge natural frequency, using a coefficient α as proposed in this paper.
2. For a typical trainload with multiple carriages, the dynamic effect is overwhelmed by the repeated passage of carriages; thus, the apparent dominant frequencies with a basic value of V/L_c , which may also be looked upon as the normalized speed with respect to the carriage length, becomes a governing factor. However, there exists a cluster of dominant frequencies at equal intervals of V/L_c , and theoretically speaking, some degree of resonance would occur when any of such dominant frequencies coincides with the bridge natural frequency. Therefore, it requires specific evaluation of the resonance severity before such a resonance condition becomes practically meaningful.
3. A resonance indicator, the Z factor, is thus developed. The Z factor represents the rate of increase of the bridge response amplitudes as the consecutive moving loads pass over the bridge at a potential resonance speed. It is found that the Z factor is essentially only dependent on the length ratio between the bridge and the carriage. The resonance effect is shown to be markedly more severe in shorter bridges (with $L_b/L_c \leq 1.5$).
4. With the Z factor, for a given bridge (effectively the ratio L_b/L_c and the natural frequency), the relative resonance severity at different orders of the dominant frequencies, which correspond to resonance speeds, can be predicted conveniently. Numerical results confirm that the prediction agrees very well with the detailed finite-element analysis results. When the carriage mass effect is involved, the Z factor can still be used effectively to predict the resonance effect, provided that an appropriate effective natural frequency is used in the establishment of the resonance speeds.

The Z factor provides a convenient means for the identification of the speeds that could result in most serious resonance effects. Such information can assist in decision making with regard to regulating

the traffic speed for existing railway bridges, especially when high-speed trains are involved. It can also be used to inform the design of new bridges such that optimization of variable design parameters, for instance, increasing the natural frequency and adjusting the bridge length where possible, may be considered to avoid serious resonances. Continued studies are being carried out to develop more specific guidelines along this line.

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